

Direct Synthesis of *n*-Butanol from Ethanol over Nonstoichiometric Hydroxyapatite

Takashi Tsuchida,^{*,†} Shuji Sakuma,[†] Tatsuya Takeguchi,[‡] and Wataru Ueda[‡]

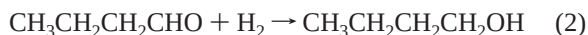
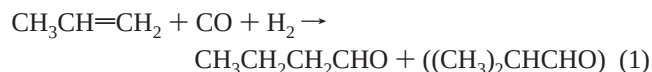
Central Research Center, Sangi Co., Ltd. Fudojino 2745-1, Kasukabe-shi, Saitama 344-0001, Japan, and
Catalysis Research Center, Hokkaido University, Kita 21 Nishi 10, Kita-ku, Sapporo 001-0021, Japan

n-Butanol is an important industrial chemical usually produced by the oxo process, an expensive, energy-consuming set of reactions over metal catalysts, using petrochemical raw materials at high pressure. We developed nonstoichiometric hydroxyapatite (HAP), a highly active calcium phosphate compound and found it catalyzed selective conversion of ethanol to *n*-butanol in a single reaction at atmospheric pressure and low temperature, with maximum selectivity of 76%. Higher alcohols were also formed. We postulate that ethanol is adsorbed and activated on HAP as CH₃CH₂OH(a) and that a C–C bond was formed between β-C in the CH₃CH₂OH(a) and α-C in *n*-C_{*n*}H_{2*n*+1}OH to produce *n*-C_{*n*}H_{2*n*+1}CH₂CH₂OH. We further postulate that, by successive propagation, part of this *n*-C_{*n*}H_{2*n*+1}CH₂CH₂OH is then adsorbed and activated on HAP as *n*-C_{*n*}H_{2*n*+1}CH₂CH₂OH(a) and that C–C bond was formed between β-C in the *n*-C_{*n*}H_{2*n*+1}CHCH₂OH(a) and α-C in *n*-alcohol to produce branched alcohols. Reaction simulation supported this hypothesis, suggesting that efficient, environmentally friendly production of *n*-butanol might be possible in future using bioethanol as raw material.

Introduction

Synthesis of *n*-Butanol. *n*-Butanol is an important chemical feedstock used as a solvent and in polymer raw materials such as butyl acrylate and butyl methacrylate. World production was estimated to be about 2.9 million tons in 2005.¹ Markets for *n*-butanol in Japan, Europe, and America have mostly peaked, but growth is still strong in developing countries, particularly Brazil, Russia, India, and China (BRICs), for use in car manufacture, building construction, etc. Demand is expected to grow steadily worldwide because of the wide range of applications for *n*-butanol in many industries.

n-Butanol can be synthesized by the oxo process,^{2–7} by the acetaldehyde method (so-called aldol condensation), or by fermentation (a bioprocess), but in the petrochemical industry the oxo process is generally used. This consists of two successive reactions: the hydroformylation of propylene with syngas (carbon monoxide and hydrogen) provided from oil on cobalt or rhodium catalysts at high pressure to produce aldehyde, and the hydrogenation of this aldehyde on nickel catalyst to provide *n*-butanol:



A large quantity of energy is required to reform natural gas (methane) at 800 °C with steam to get syngas. Given the need for “green technologies” to resolve global warming problems in the 21st century, a new form of energy-saving process for production of *n*-butanol is desirable. Furthermore, the oxo method uses harmful carbon monoxide as a raw material at high pressure in addition to propylene and is a complicated process, resulting in high cost and low profitability. Since the interna-

tional oil price exceeded \$70 per barrel in 2005, manufacturing costs of these higher alcohols have risen owing to the sudden rise in the propylene price, and profits have declined. As a result, prices of derivatives of *n*-butanol have also risen.

On the other hand, bioethanol is regarded as carbon-neutral in the Kyoto Protocol adopted at the Third Conference of the Parties to the United Nations Framework Convention on Climate Change (COP3). Production of bioethanol is increasing in Canada, China, India, Australia, and European countries, and in 2002 amounted to 33.86 × 10⁶ kL worldwide, the main producers being Brazil > USA > China > India in that order.⁸ Bioethanol from a variety of waste cellulose materials such as bagasse, rice, and wheat straw, stems and leaves of corn, wastepaper, waste wood, and garbage has been developed,^{9–12} and the quality of the bioethanol is completely the same as the commercial ethanol. Brazil is the most advanced country, with a full bioethanol business infrastructure, from the cultivation of sugar cane to the logistics and commercial use of ethanol, already in place, and a production cost estimated to be the world's cheapest at \$14–15/barrel in 2004.

Synthesis of higher alcohol from lower alcohol is generally known as the Guerbet reaction,^{13–18} in which solid-base catalysts or supported metal catalysts are used. In the case of *n*-butanol, the mechanism of synthesis from ethanol is considered to be an indirect process,^{19–22} in which ethanol is converted to *n*-butanol via acetaldehyde. However, the possibility of direct synthesis of *n*-butanol from ethanol over a multicomponent oxide catalyst (MgO–CuO–MnO),²³ over alkali cation zeolites,²⁴ or over a MgO catalyst²⁵ has also been suggested. Although the elementary reactions of this synthesis have not been clarified, the process involves dehydration of two molecules of ethanol in the overall chemical reaction formula:



Since this process is clean, it theoretically fits the “green chemistry” requirements of the 21st century. However, in the case of the MgO catalyst, selectivity to *n*-butanol was around 40%, leaving much room for improvement before industrialization becomes feasible.

* Corresponding author. Telephone: +81-48-752-0111. Fax: +81-48-752-0120. E-mail: takashi.tsuchida@sangi-j.co.jp.

[†] Sangi Co.

[‡] Hokkaido University.

In this study, the authors developed nonstoichiometric hydroxyapatite, a highly active calcium phosphate compound, and found that it catalyzed selective conversion of ethanol to *n*-butanol in a one-step process, with a maximum selectivity of 76%.

Characteristics of Hydroxyapatite. General properties of hydroxyapatite (HAP) may be found in the specialist literature and reviews.^{26–28} As the main component of the bones and teeth of vertebrates and fish, its applications in areas such as bone reconstruction, dental materials, and drug delivery systems have been actively studied. In this paper, the catalytic properties of nonstoichiometric HAP are discussed.

Among the apatite-type crystal structures, HAP is classified as a distorted apatite²⁹ based on the relationship in size between its Ca^{2+} and P^{5+} ions, which allows it to tolerate a certain degree of loss or substitution of ions during crystal formation and thus fall easily into a nonstoichiometric composition. Stoichiometric HAP ($\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$) has a Ca/P molar ratio (Ca/P ratio) of 1.67 and H_2O content of 1.79 wt %, but in nonstoichiometric forms, its Ca/P ratio can range from 1.50 to 1.70 with the loss of calcium ions.

HAP Catalysts. As a catalyst, HAP has the unusual property of containing both acid sites and base sites in a single-crystal lattice. It has been postulated that in nonstoichiometric HAP, the loss of Ca^{2+} ions results in an electrical imbalance that is corrected by both the introduction of H^+ ions and the loss of OH^- ions. Nonstoichiometric HAP is represented by the formula $\text{Ca}_{10-z}(\text{HPO}_4)_z(\text{PO}_4)_{6-z}(\text{OH})_{2-n}\cdot n\text{H}_2\text{O}$; $0 < z \leq 1$, $n = 0\text{--}2.5$, with the stipulation that OH^- sites in this formula can be partially substituted with H_2O .²⁶ Consequently, it can be expected that, within the nonstoichiometric HAP crystal, new acid sites can be created by the formation of phosphate groups in areas of lattice defects and new base sites by the attachment at other points on the lattice of structural hydroxyl or adsorbed water groups.

Many studies on the catalytic properties of HAP have been reported, falling roughly into three groups. The first, as in this report, study the catalytic properties of HAP itself.^{30–36} For example, HAP with a low Ca/P ratio showed a lower capacity to convert ethanol to ethylene or butanol to butene (dehydration processes) than zeolite or aluminum acid catalysts,^{30–33,35} whereas HAP with a higher Ca/P ratio converted alcohol to aldehyde or ketone (dehydrogenation).^{34–36} A second group of studies examines the unique function of ion-exchanged HAP catalysts.^{37–40} For example, a HAP catalyst with Ca^{2+} ions replaced by strontium showed oxidative dehydrogenation of ethane in the presence of tetrachloromethane.³⁷ Direct synthesis of phenol from benzene has also been reported, over a HAP catalyst containing both Ca^{2+} and Cu^{2+} ions in its cation portion.³⁸ A third group of studies focuses on metal catalysts supported on HAP, which show high catalytic activity in the oxidation of alcohols.⁴¹

Experimental Section

Initial Catalyst Screening. As is well-known, only ethylene can be synthesized from ethanol by dehydration on a typical solid acid catalyst like alumina. We therefore began our research by screening a wide range of base catalysts, such as MgO , and commercially available acid–base catalysts for their ability to catalyze *n*-butanol synthesis from ethanol. Catalysts screened were as follows: MgO (particle size: $0.05\text{ }\mu\text{m}$), $\text{Mg}(\text{OH})_2$ (particle size: $0.07\text{ }\mu\text{m}$), $\text{Mg}_3(\text{PO}_4)_2\cdot 8\text{H}_2\text{O}$, CaO , $\text{Ca}(\text{OH})_2$, CaF_2 , CaSiO_3 , $\text{CaSO}_4\cdot 2\text{H}_2\text{O}$, Li_3PO_4 , AlPO_4 , $\text{Ca}_{10}(\text{PO}_4)_6\text{F}_2$ (fluorapatite: FAP), $\text{Ca}_4(\text{PO}_4)_2\text{O}$ (tetracalcium phosphate: TTCP),

hydrotalcite, talc, and kaolin (all of which were purchased from Wako Pure Chemical Industries, Ltd.) and sepiolite (purchased from Mizusawa Industrial Chemicals, Ltd.). Nonstoichiometric HAP at various Ca/P ratios, prepared by the method shown in the next section, was also used.

Synthesis of HAP Catalysts. To examine the effect of the Ca/P ratio of HAP catalysts, HAP catalysts with various Ca/P ratios were prepared by the precipitation method. A mixed solution containing 0.60 mol/L of $\text{Ca}(\text{NO}_3)_2\cdot 4\text{H}_2\text{O}$ and 0.40 mol/L of $(\text{NH}_4)_2\text{HPO}_4$ was titrated with aqueous ammonia to a pH of 9–10. The solution was then stirred for 24 h and filtered. The filtrate was washed with deionized water and dried at $140\text{ }^\circ\text{C}$. The powder obtained was mixed with deionized water so that the resulting slurry contained 10 wt % powder and then further pulverized in a ball mill for 48 h. The gel obtained from this process was aged and dried at $140\text{ }^\circ\text{C}$. The resulting substance was ground in a mortar and then baked for 2 h at $600\text{ }^\circ\text{C}$ to produce the final HAP powder catalyst.

Characterization of HAP. Each catalyst was characterized by powder X-ray diffraction and measurement of its BET surface area. The Ca/P ratio for each catalyst was measured by fluorescent X-ray analysis. Powder X-ray diffraction confirmed that each catalyst synthesized was composed of crystalline HAP, and the BET surface area of each was between 30 and $50\text{ m}^2/\text{g}$.

Reaction Method. Nonstoichiometric HAP powders of a variety of Ca/P ratios were respectively placed in a molding machine to form pellets. The pellets were then lightly crushed to a particle size of 14 to 26 mesh for reaction with ethanol. Ethanol (16.4 vol %) diluted with helium was passed through a fixed-bed silica tubular reactor with interior diameter of 5 mm. Products were analyzed by gas chromatograph with mass spectrometer (GC–MS) in the range $m/z = 10\text{--}400$ and by gas chromatograph with flame ionization detector (GC–FID). In both cases, columns supplied by J&W Scientific Corporation (liquid phase: DB-1, film thickness: $5.00\text{ }\mu\text{m}$, column dimension: $30\text{ m} \times 0.323\text{ mm}$) were used. Note that GC–MS showed the total amount of CO and CO_2 in the reaction product throughout the experiment was less than 1.0%, while methane formation/carbon deposition on the catalyst was negligible, indicating that ethanol was not decomposed on HAP catalysts of varying Ca/P ratios over the range of temperatures used in this study.

Calculation Formulas. The ethanol conversion, yield, and selectivity of products were calculated as follows:

$$\text{ethanol conversion (\%)} = \frac{(1 - \text{C mol of unreacted ethanol})}{\text{C mol of total outlet gases}} \times 100$$

$$\text{product yield (\%)} = \frac{\text{C mol of product}}{\text{C mol of total outlet gases}} \times 100$$

$$\text{product selectivity (\%)} = \frac{\text{C mol of product}}{\text{C mol of reacted ethanol}} \times 100$$

Results and Discussion

***n*-Butanol Synthesis from Ethanol on Various Catalysts.** Table 1 shows the performance of the various catalysts in promoting *n*-butanol synthesis from ethanol. The highest yields and respective conversion reaction temperatures are shown. Nonstoichiometric HAP (Ca/P ratio 1.61) exhibited the highest *n*-butanol yield at 19.8%, followed by hydrotalcite at 12.2%, $\text{Mg}(\text{OH})_2$ (particle size: $0.07\text{ }\mu\text{m}$) at 9.0%, and CaF_2 at 5.2%. While *n*-butanol was formed on catalysts with OH groups and

Table 1. Yield of *n*-Butanol Synthesis from Ethanol on Various Catalysts

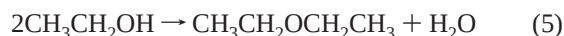
entry	catalyst	yield/%	temp/°C
1	MgO (0.05 μ m)	4.9	500
2	Mg(OH) ₂ (0.07 μ m)	9.0	450
3	Mg ₃ (PO ₄) ₂ ·8H ₂ O	tr	
4	CaO	1.2	450
5	Ca(OH) ₂	2.2	450
6	CaF ₂	5.2	500
7	CaSiO ₃	2.1	450
8	CaSO ₄ ·2H ₂ O	1.3	500
9	Li ₃ PO ₄	tr	
10	AlPO ₄	tr	
11	hydroxyapatite (HAP; Ca/P ratio 1.61)	19.8	350
12	Ca ₁₀ (PO ₄) ₆ F ₂ (FAP)	1.0	400
13	Ca ₄ (PO ₄) ₂ O (TTCP)	3.0	500
14	hydrotalcite	12.2	350
15	sepiolite	tr	350
16	talc	tr	
17	kaolin	tr	

the basic oxides, it was not formed on Mg₃(PO₄)₂·8H₂O, Li₃PO₄, talc, kaolin, or AlPO₄.

Relation between Ca/P Ratio of HAP Catalysts and Reaction Products. The effect of the HAP catalyst's Ca/P ratio on ethanol conversion is shown in Figure 1, under the following conditions: feed gas, 20 vol % ethanol/He; gas space velocity, 10 000 L/h; catalyst volume, 0.6 mL; reaction temperature, 400 °C. HAP catalyst with a low Ca/P ratio of 1.52 converted ethanol to ethylene with 80% selectivity by dehydration. This result agreed with that of Monma.³⁵ However, HAP catalysts with low Ca/P ratios also produced a small amount of acetaldehyde, which is not synthesized on solid acid catalysts such as Al₂O₃ or zeolite. We postulated that HAP catalysts with low Ca/P ratios have both a large number of acid sites and a small number of base sites and that HAP with predominantly acid sites functions as a dehydration catalyst:



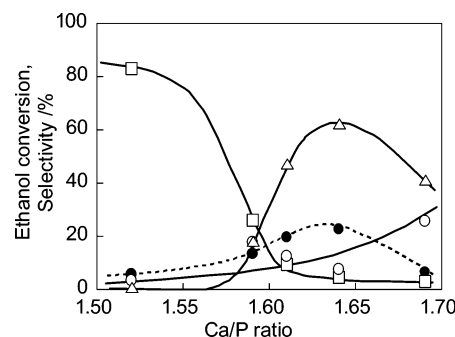
Our experiments also showed that ethanol is dimerized to diethyl ether on HAP catalysts with low Ca/P ratios at low temperatures (data not shown—refer to Table 2), which supports the hypothesis that these catalysts contain a large number of acid sites:



On the other hand, HAP catalysts with higher Ca/P ratios, closer to stoichiometry, produced *n*-butanol with the highest selectivity among the products and acetaldehyde with next highest selectivity. In the case of acetaldehyde, selectivity was highest (24.9%) at a Ca/P ratio of 1.69. The results indicated that selectivity to acetaldehyde increases with an increase in the HAP catalyst's Ca/P ratio and also agreed with those of Monma.³⁵ We postulate that HAP catalysts with high Ca/P ratios have a small number of acid sites and a large number of base sites and that HAP with predominantly base sites functions as a dehydrogenation catalyst. This hypothesis is supported by the fact that ethanol is generally converted to acetaldehyde on a solid base catalyst:



HAP catalyst with a Ca/P ratio of 1.64 showed both the highest conversion of ethanol (22.7%) and the highest selectivity to *n*-butanol (62.4%). These data suggest that HAP with a Ca/P ratio of 1.64 has a large number of both acid sites and base sites and that pairs of acid sites and base sites play an important role in the activation of ethanol to produce *n*-butanol. We

**Figure 1.** Relationship of Ca/P ratio of HAP catalysts to ethanol conversion (shown as ●) and selectivities of products. Selectivity to *n*-butanol, ethylene, and acetaldehyde are shown as Δ, □, and ○, respectively. Feed gas, 20% ethanol–80% He; GHSV, 10 000 h⁻¹; reaction temperature, 400 °C.**Table 2.** Effect of Temperature and Contact Time on Product Selectivity from Ethanol on HAP (Ca/P ratio = 1.64)

reaction temp (°C)	300	300	300	300	350	400	450
contact time/s	0.02	0.45	0.89	1.78	1.78	1.78	1.78
conversion (%)	0.6	4.2	10.7	14.7	26.1	57.4	95.3
Selectivity (C wt %)							
CH ₄	0.7	0.1	tr	tr	tr	tr	0.2
C ₂ H ₄	13.5	2.5	0.7	0.6	1.6	3.6	6.1
C ₂ H ₆	10.0	tr	0.2	tr	tr	tr	tr
CH ₃ CHO	61.7	11.5	4.4	1.7	3.0	2.3	1.3
C ₃ H ₆	0.3	0.1	0.1	tr	0.2	0.5	1.5
CH ₃ COCH ₃	tr	tr	tr	tr	tr	tr	0.1
CH ₃ CH ₂ CH ₂ OH	tr	tr	tr	tr	tr	tr	0.1
C ₄ H ₈	2.4	2.0	1.2	0.4	0.8	2.3	6.2
CH ₂ =CH–CH=CH ₂	3.3	3.0	1.8	1.1	2.3	5.9	11.7
C ₃ H ₈ CHO	tr	0.4	0.4	0.2	0.7	1.3	1.4
CH ₃ CH=CHCHO	tr	0.4	0.2	0.1	0.2	0.1	tr
C ₂ H ₅ OC ₂ H ₅	tr	tr	0.1	0.1	0.4	0.4	0.4
<i>n</i> -C ₄ H ₉ OH	3.2	61.4	75.0	76.3	68.8	44.8	6.0
C ₆ -alcohol	0.3	5.0	6.2	8.6	10.1	13.7	5.6
C ₈ -alcohol	tr	0.3	0.5	1.0	1.6	4.1	3.3
C ₁₀ -alcohol	0.5	tr	0.1	0.2	0.3	1.0	1.4
aromatics	tr	1.0	1.0	1.3	1.7	3.0	11.2
others	4.1	12.2	8.1	8.5	8.4	16.8	43.4

concluded that HAP with a Ca/P ratio of 1.64 was the most suitable catalyst to use for selective *n*-butanol synthesis.

Reaction of Ethanol on HAP Catalyst with a Ca/P Ratio of 1.64

Analysis of Reaction Products. Table 2 shows the product distribution from conversion of ethanol on HAP with a Ca/P ratio of 1.64 at various temperatures and contact times. Only a trace of the C₁ product methane was formed, indicating that ethanol was not decomposed on the catalyst at temperatures of 300–450 °C. C₂ products were mainly ethylene and acetaldehyde, suggesting that ethylene was formed by dehydration of ethanol on the catalyst's acid sites and acetaldehyde by dehydrogenation of ethanol on its base sites. Ethylene selectivity at a fixed contact time of 1.78 s was seen to rise with a rise in temperature, suggesting that the postulated function of HAP as predominantly an acid catalyst at a Ca/P ratio of 1.64 is enhanced at high temperature. On the other hand, selectivity to acetaldehyde, which we postulated to be formed on the base sites of the HAP catalyst, tended to decrease with a rise in temperature above 350 °C at a fixed contact time of 1.78 s, suggesting that acetaldehyde is a reaction intermediate. Only very small amounts of the C₃ products propylene, acetone, and 1-propanol were detected. This suggests that propylene synthesis resulted from the oligomerization and subsequent decomposition of olefins.

In the case of C₄ products, *n*-butanol, 1,3-butadiene, butene, unsaturated alcohols, butyraldehyde, and diethyl ether were

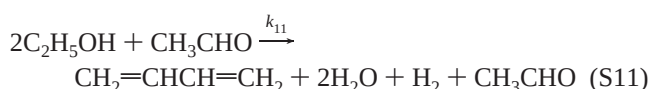
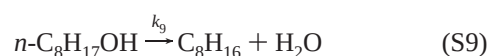
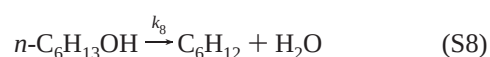
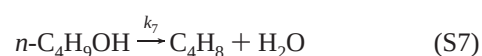
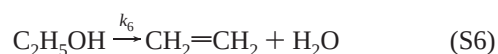
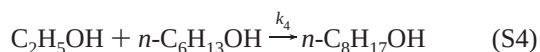
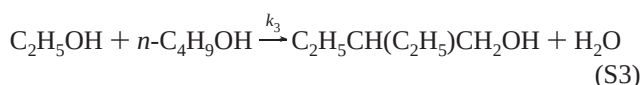
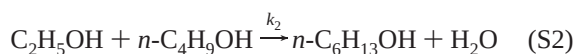
formed. Selectivity to *n*-butanol reached its maximum value of 76.3% at 300 °C, contact time 1.78 s, which to the best of our knowledge is the highest selectivity to *n*-butanol ever reported. Thereafter it decreased with a rise in reaction temperature, whereas from 300 to 400 °C, selectivity to C₆ alcohols increased, suggesting that over this temperature range *n*-butanol reacted with ethanol to form C₆ alcohol. Selectivity to butene and butyraldehyde also rose with a rise in reaction temperature, suggesting that at higher temperatures *n*-butanol was dehydrated to butene on the acid sites of the HAP catalyst and that it was dehydrogenated to butyraldehyde on the base sites of the catalyst. We concluded that a reaction temperature of 300 °C is most suitable for synthesis of *n*-butanol, as our results indicate that at 300 °C almost no reaction between ethanol and *n*-butanol occurs and dehydration and dehydrogenation of *n*-butanol on the HAP catalyst are also minimal.

C₅ products and above included alcohols such as 2-ethyl-1-butanol; *n*-hexanol; 2-ethyl-1-hexanol; *n*-octanol; two kinds of branched C₁₀ alcohols; *n*-decanol; C₆ and C₈ unsaturated alcohols; C₅, C₆, C₈, and C₁₀ olefins, dienes, and trienes and the aromatics benzene, toluene, xylene, benzaldehyde, and butylbenzene. With an increase in reaction temperature from 400 to 450 °C, selectivity to all alcohols except C₁₀ alcohols considerably decreased, while selectivity to aromatics and other hydrocarbons considerably increased, indicating that on HAP with a Ca/P ratio of 1.64, alcohols are converted to aromatics and other hydrocarbons at high temperature.

Effect of Contact Time on Synthesis of Alcohols. The effect of contact time on product selectivity is also shown in Table 2. Alcohols produced were *n*-butanol; C₆ alcohols comprising 2-ethyl-1-butanol and *n*-hexanol; C₈ alcohols comprising 2-ethyl-1-hexanol and *n*-octanol; and C₁₀ alcohols comprising 2-ethyl-1-octanol, 2-butyl-1-hexanol, and *n*-decanol. No other isomeric alcohols were observed. These results indicate that even-carbon-numbered alcohols are selectively synthesized from ethanol on HAP catalyst with a Ca/P ratio of 1.64. Moreover, in contrast with methane, ethane, ethylene, propylene, and acetaldehyde, whose selectivity decreased with an increase in contact time, selectivity to C₆, C₈, and C₁₀ alcohols, while still low level, was seen to increase with an increase in contact time, suggesting that successive propagation of alcohols occurred. In the case of *n*-butanol, at a fixed temperature of 300 °C, selectivity was only 3.2% at a contact time of 0.02 s but rose to its highest level of 76.3% at a contact time of 1.78 s. As noted earlier, ethanol has been converted to *n*-butanol on MgO catalyst with selectivity of around 40%, but successive propagation to higher alcohols was not reported.²⁵ To the best of our knowledge, this paper is the first to report successive propagation of ethanol.

Effect of Reaction Temperature on Synthesis of Alcohols. The effect of reaction temperature on yields of various alcohols from ethanol on HAP with a Ca/P ratio of 1.64 is shown in Figure 2. Maximum yields of C₄ alcohol and C₁₀ alcohols were obtained at 400 and 450 °C, respectively, suggesting that the temperature at which maximum yields were obtained increased with an increase in the alcohol's C number. These data also suggest a successive propagation mechanism. Yields of aromatics and other hydrocarbons (data not shown—refer to Table 2) also increased with an increase in reaction temperature, suggesting that alcohols were converted to aromatics and other hydrocarbons by dehydration or dehydrogenation polymerization on the HAP catalyst at higher temperatures. The authors' findings and conclusions regarding the effect of reaction temperature on selectivity for *n*-butanol are as presented in the section Analysis of Reaction Products.

Proposed Reaction Steps of *n*-Butanol and Other Byproduct Synthesis on HAP. Our results showed that hydroxyapatite with a Ca/P ratio of 1.64 catalyzed selective conversion of ethanol to *n*-butanol at 300 °C, contact time 1.78 s, with a maximum selectivity of 76.3%, in a process in which higher alcohols and other hydrocarbon byproducts were formed. However, the mechanism of this reaction process has not been elucidated. Assuming that only reactions between normal alcohols (*n*-alcohols) occurred, the authors propose the following reaction steps:



On the basis of the above reaction steps (S1–S13), the reaction rate for each product can be expressed as follows, where reaction constants k_1 – k_{13} are those for reaction steps S1–S13, respectively. Those reaction steps did not conflict with the well-accepted Guerbet reaction:^{13–18}

$$\begin{aligned} d[\text{C}_2\text{H}_5\text{OH}]/dt = & -2k_1[\text{C}_2\text{H}_5\text{OH}]^2 - \\ & (k_2 + k_3)[\text{C}_2\text{H}_5\text{OH}][n\text{-C}_4\text{H}_9\text{OH}] - \\ & (k_4 + k_5)[\text{C}_2\text{H}_5\text{OH}][n\text{-C}_6\text{H}_{13}\text{OH}] - \\ & (k_6 + k_{11} + k_{13})[\text{C}_2\text{H}_5\text{OH}] - 2k_{10}[\text{C}_2\text{H}_5\text{OH}][\text{CH}_3\text{CHO}] \end{aligned} \quad (\text{R0})$$

$$\begin{aligned} d[n\text{-C}_4\text{H}_9\text{OH}]/dt = & k_1[\text{C}_2\text{H}_5\text{OH}]^2 - \\ & (k_2 + k_3)[\text{C}_2\text{H}_5\text{OH}][n\text{-C}_4\text{H}_9\text{OH}] - k_7[n\text{-C}_4\text{H}_9\text{OH}] \end{aligned} \quad (\text{R1})$$

$$d[n\text{-C}_6\text{H}_{13}\text{OH}]/dt = k_2[\text{C}_2\text{H}_5\text{OH}][n\text{-C}_4\text{H}_9\text{OH}] - (k_4 + k_5)[\text{C}_2\text{H}_5\text{OH}][n\text{-C}_6\text{H}_{13}\text{OH}] - k_8[n\text{-C}_6\text{H}_{13}\text{OH}] \quad (\text{R2})$$

$$d[\text{C}_2\text{H}_5\text{CH}(\text{C}_2\text{H}_5)\text{CH}_2\text{OH}]/dt = k_3[\text{C}_2\text{H}_5\text{OH}][n\text{-C}_4\text{H}_9\text{OH}] \quad (\text{R3})$$

$$d[n\text{-C}_8\text{H}_{17}\text{OH}]/dt = k_4[\text{C}_2\text{H}_5\text{OH}][n\text{-C}_6\text{H}_{13}\text{OH}] - k_9[n\text{-C}_8\text{H}_{17}\text{OH}] \quad (\text{R4})$$

$$d[\text{C}_4\text{H}_9\text{CH}(\text{C}_2\text{H}_5)\text{CH}_2\text{OH}]/dt = k_5[\text{C}_2\text{H}_5\text{OH}][n\text{-C}_6\text{H}_{13}\text{OH}] \quad (\text{R5})$$

$$d[\text{C}_2\text{H}_4]/dt = k_6[\text{C}_2\text{H}_5\text{OH}] \quad (\text{R6})$$

$$d[\text{C}_4\text{H}_8]/dt = k_7[n\text{-C}_4\text{H}_9\text{OH}] \quad (\text{R7})$$

$$d[\text{C}_6\text{H}_{12}]/dt = k_8[n\text{-C}_6\text{H}_{13}\text{OH}] \quad (\text{R8})$$

$$d[\text{C}_8\text{H}_{16}]/dt = k_9[n\text{-C}_8\text{H}_{17}\text{OH}] \quad (\text{R9})$$

$$d[\text{CH}_3\text{CHO}]/dt = k_{10}[\text{C}_2\text{H}_5\text{OH}] \quad (\text{R10})$$

$$d[\text{CH}_2=\text{CHCH}=\text{CH}_2]/dt = k_{11}[\text{C}_2\text{H}_5\text{OH}][\text{CH}_3\text{CHO}] \quad (\text{R11})$$

$\text{CH}_2=\text{CHCH}=\text{CH}_2$ can be expected to be formed by the Lebedev reaction, in which case its formation rate can be considered proportional to $[\text{C}_2\text{H}_5\text{OH}][\text{CH}_3\text{CHO}]$, with CH_3CHO functioning as catalyst. This will be discussed later:

$$d[\text{aromatics}]/dt = k_{12}[\text{C}_2\text{H}_5\text{OH}] \quad (\text{R12})$$

$$d[\text{others}]/dt = k_{13}[\text{C}_2\text{H}_5\text{OH}] \quad (\text{R13})$$

Aromatics may be formed by various routes from various alcohols. To simplify reactions for simulation purposes, the total rate of formation of aromatics was assumed to be proportional to the total ethanol concentration during their formation period, and aromatics concentration is expressed in terms of C_6 aromatics (R12). Similarly, the total rate of formation of other hydrocarbons was assumed to be proportional to the total ethanol concentration during their formation period, and concentration for this group is expressed in terms of C_8 hydrocarbons (R13).

Normal and Branched Alcohols. To clarify the reaction mechanism in the case of n -alcohols, the effect of contact time on yields of n -alcohols was examined at 450 °C. Results are shown in Figure 3. Lines show simulation curves for the ethanol conversion rate and yields of n -alcohols based on the above reaction rates (R0–R13), using the parameters shown in Table 3.

As seen in Figure 3, the simulation curves fit the reaction data well. The n -butanol yield increases with an increase in contact time from 0 to 0.22 s, with the HAP catalyst showing its maximum n -butanol yield of 19.3% at 0.22 s. The n -butanol yield thereafter decreases with an increase in contact time, whereas the n -hexanol yield increases with an increase in contact time from 0 to 0.67 s, with the HAP catalyst showing its maximum n -hexanol yield of 3.8% at 0.67 s. The n -hexanol yield thereafter decreases with an increase in contact time over 0.67 s (data not shown). Data for n -octanol follows a similar pattern. This tendency in product yields was the same at 500 °C (data not shown) as at 450 °C, suggesting the activity of a successive propagation at high temperature. Since the contact time at which maximum yields were obtained increased, respectively, in order of their C number, for n -butanol, n -hexanol, n -octanol, and n -decanol, all of which have an even

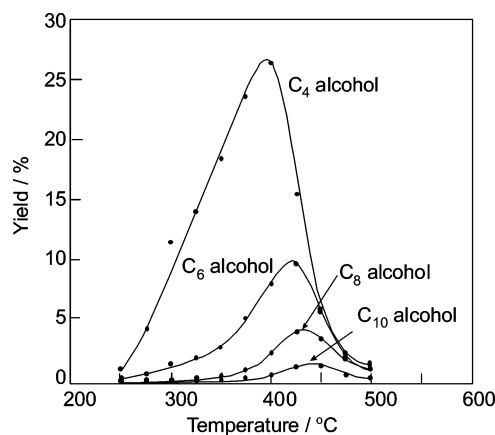


Figure 2. Effect of reaction temperature on yields of various alcohols from ethanol on HAP (Ca/P ratio = 1.64), contact time 1.78 s.

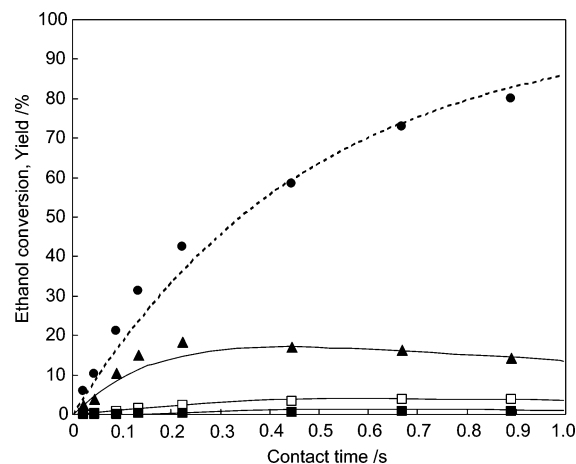
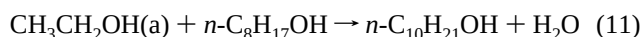
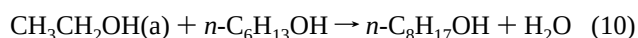
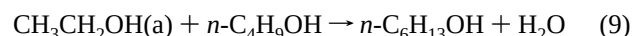
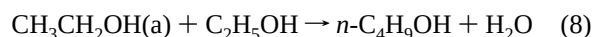


Figure 3. Simulation curves vs actual data showing the effect of contact time on ethanol conversion rate (shown as ●) and yields of normal alcohols at high temperature (450 °C), (HAP catalyst Ca/P ratio = 1.64). Yields of n -butanol, n -hexanol, and n -octanol are shown as ▲, □, and ■, respectively. Lines are simulation curves for the experimental data in each case.

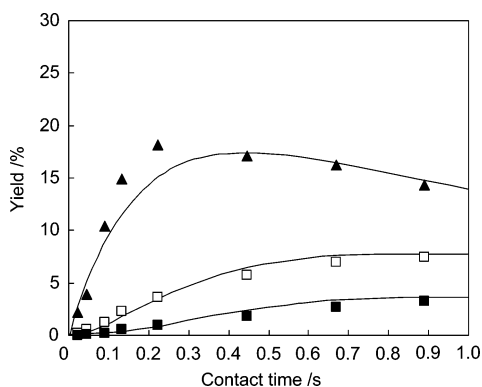
carbon number, we postulated that n -alcohols were formed via the reactions below, in the following successive propagation steps. First, ethanol was adsorbed and activated on HAP as $\text{CH}_3\text{CH}_2\text{OH}(\text{a})$. Second, the C–C bond was formed between the β -C in $\text{CH}_3\text{CH}_2\text{OH}(\text{a})$ and α -C in $n\text{-C}_n\text{H}_{2n+1}\text{OH}$ to form $n\text{-C}_{n+2}\text{H}_{2n+5}\text{OH}$. These reaction steps did not conflict with well-accepted Guerbet reaction containing aldol condensation mechanism.^{13–18} To clarify the real reaction mechanism, further detail investigations using isotope compounds will be necessarily. Therefore, apparent reaction steps will be discussed in this paper:



The effect of contact time on yields of total alcohols, both normal and branched, at 450 °C is shown in Figure 4. In the case of C_4 alcohol, however, only n -butanol was obtained. The yield of normal and branched C_6 alcohols, comprising n -hexanol and 2-ethyl-1-butanol, increased with an increase in contact time

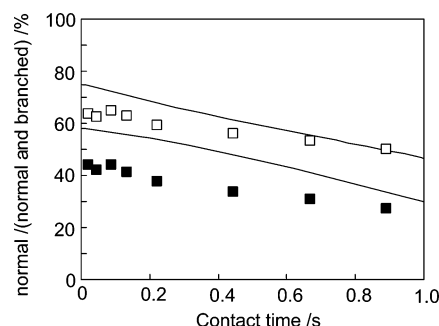
Table 3. Reaction Rate Constants for *n*-Butanol Synthesis from Ethanol on HAP (Ca/P = 1.64) Catalyst at 300, 350, 400, and 450 °C, Showing Activation Energy (E_a) and Frequency Factor for Each Reaction Step^a

reaction step		reaction rate constant (k_i /unit)				A /unit	unit	E_a (kJ/mol)
		300 °C	350 °C	400 °C	450 °C			
S1	k_1	6.1	1.3×10^1	3.3×10^1	8.8×10^1	2.1×10^6	$\text{L mol}^{-1} \text{s}^{-1}$	61
S2	k_2	8.5	1.5×10^1	4.1×10^1	2.0×10^2	1.7×10^7	$\text{L mol}^{-1} \text{s}^{-1}$	70
S3	k_3	7.2	9.6	2.1×10^1	6.5×10^1	2.0×10^5	$\text{L mol}^{-1} \text{s}^{-1}$	50
S4	k_4	5.2	2.2×10^1	5.4×10^1	3.5×10^2	1.2×10^9	$\text{L mol}^{-1} \text{s}^{-1}$	92
S5	k_5	1.1×10^2	2.9×10^1	7.9×10^1	2.4×10^2	4.2×10^3	$\text{L mol}^{-1} \text{s}^{-1}$	21
S6	k_6	5.9×10^{-4}	3.3×10^{-3}	2.0×10^{-2}	1.0×10^{-1}	3.4×10^7	s^{-1}	119
S7	k_7	8.7×10^{-3}	1.6×10^{-2}	6.9×10^{-2}	4.6×10^{-1}	1.0×10^6	s^{-1}	91
S8	k_8	2.2×10^{-1}	9.3×10^{-2}	2.8×10^{-1}	1.2	5.5×10^2	s^{-1}	40
S9	k_9	1.0	2.9×10^{-1}	7.2×10^{-1}	3.7	2.7×10^2	s^{-1}	30
S10	k_{10}	2.3×10^{-3}	8.1×10^{-3}	4.1×10^{-2}	2.0×10^{-1}	3.8×10^6	s^{-1}	102
S11	k_{11}	7.4×10^1	7.8×10^1	1.9×10^2	6.4×10^2	1.6×10^6	$\text{L mol}^{-1} \text{s}^{-1}$	49
S12	k_{12}	4.1×10^{-4}	1.2×10^{-3}	4.8×10^{-3}	4.4×10^{-2}	1.1×10^6	s^{-1}	105
S13	k_{13}	1.4×10^{-3}	2.9×10^{-3}	1.4×10^{-2}	7.5×10^{-2}	2.1×10^5	s^{-1}	91

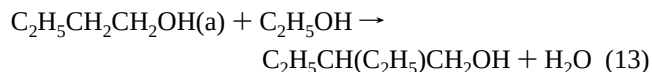
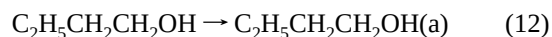
^a $[\text{C}_2\text{H}_5\text{OH}]_0 = 7.32 \text{ mmol/L}$.**Figure 4.** Simulation curves vs actual data showing the effect of contact time on yields of various alcohols, both normal and branched, at high temperature (450 °C) from ethanol on HAP (Ca/P ratio = 1.64). Yields of *n*-butanol, total (normal and branched) hexanols, and total (normal and branched) octanols are shown as $\blacktriangle$, $\square$, and $\bullet$, respectively. Lines are simulation curves for the experimental data in each case.

from 0 to 0.89 s, with the HAP catalyst showing its maximum C_6 alcohol yield of 7% at 0.89s. Total C_6 alcohol yield decreased thereafter with an increase in contact time over 0.89 s (data not shown). The yield of normal and branched C_8 alcohols showed a similar pattern. As noted also in relation to Figure 3, the contact time at which maximum yields were obtained increased respectively in the order of the alcohol's C number (data for octanols not shown). Regarding the reaction mechanism, if only $\text{CH}_3\text{CH}_2\text{OH}$ was adsorbed and activated on HAP as $\text{CH}_3\text{CH}_2\text{OH(a)}$ to form a C—C bond between β -C in $\text{CH}_3\text{CH}_2\text{OH(a)}$ and α -C in $n\text{-C}_n\text{H}_{2n+1}\text{OH}$, only *n*-alcohol could be expected to be formed. The fact that branched alcohols were also formed implies that $n\text{-C}_n\text{H}_{2n+1}\text{CH}_2\text{CH}_2\text{OH}$ is also adsorbed and activated on HAP at longer contact times and that the C—C bond was formed between β -C in $n\text{-C}_n\text{H}_{2n+1}\text{CH}_2\text{CH}_2\text{OH(a)}$ and α -C in *n*-alcohol.

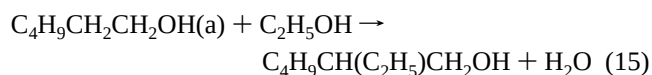
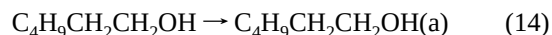
Figure 5 shows the effect of contact time on *n*-alcohol selectivity (i.e., on the ratio of normal to total (normal and branched) alcohols obtained). *n*-Alcohol selectivity for C_8 alcohols was lower than for C_6 alcohols. Although there is misfit of real and simulated data, the real data showed the same tendency as the simulation one. We postulated that, with an increase in carbon number, the mathematical combination of possible synthesis routes increases, so that selectivity for branched alcohol increases with an increase in carbon number. *n*-Alcohol selectivity also decreased with an increase in contact time, suggesting that only *n*-alcohols were activated and reacted with *n*-alcohols while branched alcohols were not activated and did not react with alcohols.

**Figure 5.** Simulation curves vs actual data showing the effect of contact time on the ratio of normal to total (normal and branched) hexanols (shown as $\square$) and the ratio of normal to total (normal and branched) octanols (shown as $\bullet$) synthesized from ethanol on HAP (Ca/P ratio = 1.64) at high temperature (450 °C). Lines are simulations of the respective experimental data in each case.

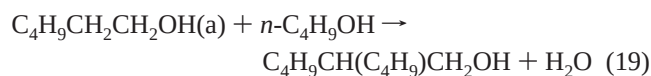
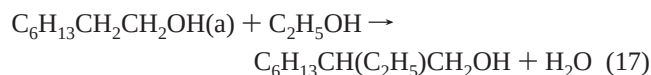
C_6 alcohols formed comprised *n*-hexanol and 2-ethyl-1-butanol. We postulate that $n\text{-C}_4\text{H}_9\text{OH}$ was adsorbed and activated on HAP as $\text{C}_2\text{H}_5\text{CH}_2\text{CH}_2\text{OH(a)}$ and that the C—C bond was formed between β -C in $\text{C}_2\text{H}_5\text{CH}_2\text{CH}_2\text{OH(a)}$ and α -C in ethanol to produce 2-ethyl-1-butanol but that no further reaction of the formed 2-ethyl-1-butanol to form a higher alcohol occurred:



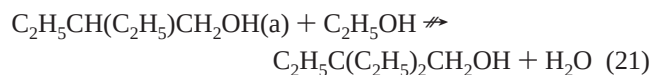
C_8 alcohols comprised 1-octanol and 2-ethyl-1-hexanol, while C_{10} alcohols comprised 1-decanol, 2-ethyl-1-octanol, and 2-butyl-1-hexanol. No other isomeric alcohols were observed. We postulate that $n\text{-C}_6\text{H}_{13}\text{OH}$ was adsorbed and activated on HAP as $\text{C}_4\text{H}_9\text{CH}_2\text{CH}_2\text{OH(a)}$ and that the C—C bond was formed between β -C in $\text{C}_4\text{H}_9\text{CH}_2\text{CH}_2\text{OH(a)}$ and α -C in ethanol to produce 2-ethyl-1-hexanol but that no further reaction of the formed 2-ethyl-1-hexanol occurred:



For C_{10} alcohol formation, we postulate the following mechanism:



If the branched alcohol formed were to react with ethanol, the following result could be expected:



The absence of the above product suggests that only *n*-alcohols—not branched alcohols—are adsorbed and activated on HAP and supports our assumption that, in the process of higher alcohol formation by successive propagation during *n*-butanol synthesis from ethanol over HAP, only reactions between *n*-alcohols occur.

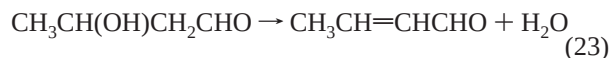
1,3-Butadiene. On the other hand, synthesis of 1,3-butadiene from ethanol is known to occur through the Lebedev reaction, for which acid–base catalysts are usually used:^{42–45}



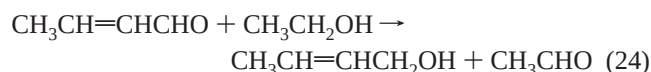
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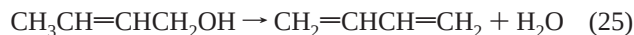
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(acid–base site)



(acid site)

The Lebedev reaction below is the overall summation of eqs 6 and 22–25:



In this reaction, CH_3CHO functions clearly as a catalyst, so that the formation rate of $\text{CH}_2=\text{CHCH}=\text{CH}_2$ can be considered proportional to $[\text{C}_2\text{H}_5\text{OH}][\text{CH}_3\text{CHO}]$. Figure 6 shows the effect of contact time on the yield of 1,3-butadiene, aldehydes, and olefins from ethanol on HAP catalyst (Ca/P ratio = 1.64) at 450 °C. Although there are some misfits for 1,3-butadiene owing to the ignorance of reaction of 1,3-butadiene, simulation curves fit the yield data for aldehydes and 1,3-butadiene well, as the figure shows. This indicates that 1,3-butadiene is formed on HAP by the Lebedev reaction. We believe our study is the first to report 1,3-butadiene synthesis from ethanol on a HAP catalyst.

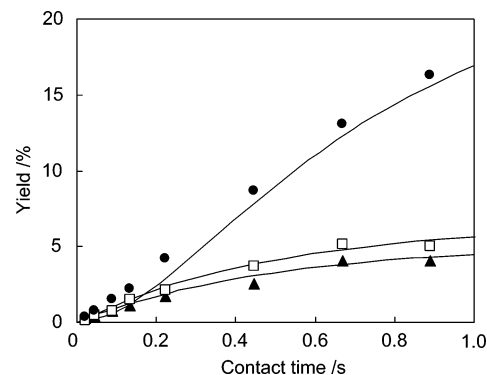
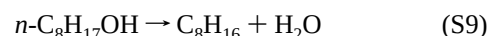
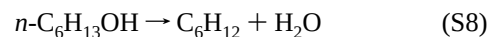
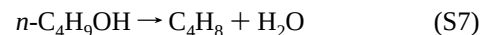


Figure 6. Simulation curves vs actual data showing the effect of contact time on the yields of 1,3-butadiene, aldehydes, and ethylene from ethanol on HAP (Ca/P ratio = 1.64) at 450 °C. Yields of 1,3-butadiene, aldehydes, and ethylene are shown as ●, □, and ▲, respectively. Lines are simulation curves for the experimental data in each case.

Olefins. Figure 6 also shows that the simulation curve for olefins fits the yield data well. If the olefins produced were formed by oligomerization of ethylene and subsequent decomposition of oligomers, formation of odd-carbon-numbered olefins such as propylene, pentene, and heptene could be expected. However, only even-carbon-numbered olefins were produced, indicating that they were derived from C_6 , C_8 , and C_{10} alcohols. As shown in Figure 5, selectivity to *n*-alcohol decreased with an increase in contact time, which also supports the view that the olefins produced derived from *n*- C_6 , *n*- C_8 , and *n*- C_{10} alcohols:



Selectivity to *n*-Butanol. Reaction rate constants for the various proposed reaction steps were determined for the range of temperatures used, and the results are shown in Table 3. Activation energy was determined by Arrhenius plots. *n*-Butanol selectivity strongly depends on $k_2 + k_3$ and $(k_2 + k_3)/k_1$, and these values were small at 300 °C but increased with an increase in reaction temperature since the activation energy of k_2 was higher than that of either k_1 or k_3 . The formation of *n*- $\text{C}_6\text{H}_{13}\text{OH}$ was strongly enhanced by an increase in reaction temperature, suggesting that the activation of *n*- $\text{C}_4\text{H}_9\text{OH}$ also plays an important role in the reaction process. At low temperature, the relative reactivity of *n*-butanol was low, so that selectivity to *n*-butanol reached over 70%, the highest ever reported. Other industrially useful higher alcohols were also selectively produced with combined selectivity for C_4 , C_6 , and C_8 alcohols totaling over 85%. Simulation results supported our finding that selectivity to *n*-butanol can be enhanced by reducing the reaction temperature.

Conclusion

n-Butanol can be synthesized from ethanol with high selectivity on HAP catalyst with a Ca/P molar ratio of 1.64. Simulation results indicate that *n*-butanol synthesis over HAP is a second-order reaction of ethanol, in which *n*-butanol is synthesized by dehydration of two molecules of ethanol. We postulated that ethanol was adsorbed and activated on HAP as $\text{CH}_3\text{CH}_2\text{OH(a)}$. HAP catalyst with a Ca/P molar ratio of 1.64 showed the highest

n-butanol selectivity of 76.3% at a contact time of 1.78 s, while reaction simulation supported our experimental finding that low temperature (300 °C) was most suitable for selective *n*-butanol synthesis from ethanol over HAP.

Higher alcohols were also synthesized on HAP from ethanol and other *n*-alcohols, by successive propagation, with combined selectivity to C₄, C₆, and C₈ alcohols exceeding 85%. The contact time at which maximum yield was obtained increased with the alcohol's carbon number. Only normal alcohol reacted with normal alcohol to form branched alcohols through a process presumed to involve C–C bond formation between β -C in *n*-C_nH_{2n+1}CH₂CH₂OH (a) and α -C in *n*-C_mH_{2m+1}CH₂CH₂OH.

Global environmental concerns along with declining production costs have drawn attention to carbon-neutral bioethanol as an alternative to petrochemical raw materials, raising the possibility that *n*-butanol synthesis from bioethanol over HAP may be a more efficient, economical, and environmentally acceptable process than the conventional oxo method.

Acknowledgment

We thank the New Energy and Industrial Technology Development Organization (NEDO) of Japan, which provided financial assistance for this research.

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Received for review May 17, 2006

Revised manuscript received September 1, 2006

Accepted September 28, 2006

IE0606082